Shor's algorithm TEMPORAL NAME

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INTRODUCTION

DAVID: Quantum advantage refers to the ability of quantum devices to outperform classical systems in solving certain problems. Although the term itself has gained popularity only recently, Richard Feynman already suggested in 1986 that exploiting quantum mechanics could enable more powerful computation than its classical counterpart [7]. In the early 1990s, several scientists, including Deutsch, Jozsa, Berthiaume, and Brassard, explicitly addressed this idea, showing that while certain tasks could be solved exactly using quantum mechanics, their classical analogues required randomization and could only be solved with high probability [4–6]. However, these works did not provide a problem outside the class of those already known to be solvable in polynomial time on a classical computer with the aid of randomization and small error [13]. In 1994, inspired by Bernstein and Vazirani's work [3], Daniel Simon made a groundbreaking contribution: he identified an oracle problem solvable in polynomial time on a quantum computer but requiring exponential time classically [12], thereby providing a concrete confirmation of Feynman's intuition.

The possibility of quantum computing excelling beyond classical one is a looming threat to, among other fields, cryptography, as some of its methods are often relied on hard-to-solve computational problems. This is the case for the widely used RSA public-key scheme developed by Rivest, Shamir and Adleman in 1978 [10]. In this system, the security rests on the difficulty of factorizing an integer into prime numbers. In 1995, the fastest known algorithm for factorizing an integer was Number Field Sieve by Lenstra, Lenstra Jr., Manasse and Pollard [9]. Such method takes an asymptotic running time $O(exp(c(\log n)^{\frac{1}{3}}(\log\log n)^{\frac{2}{3}}))$ to factor an integer n and for some constant c.

In this article, Shor introduces the first quantum algorithm for integer factorization with an asymptotic running time of $O((\log n)^2(\log\log n)(\log\log\log n))$, demonstrating a clear quantum advantage over Lenstra's classical proposal.

MATHEMATICAL BACKGROUND OF THE ALGORITHM

In this section, we address how to reduce the factorization problem to a modular exponentiation problem. SERGIO: It is not exactly that... we present the factorization procedure and we see that the bottleneck is the modular exponentiation, which can be solved using quantum computing.

SERGIO: The goal of the algorithm is the following: given a positive integer n, find $m, t \in \mathbb{N}$, where $m, t > 1$, such that $n = mt$, that is, to decompose n as a product of two non-trivial positive integers. It has been later showed that n can be assumed to be neither odd or a prime power, as for the first case 2 is a trivial factor and an efficient classical algorithm can be found for the latter[1, 2]. SERGIO: I would remove the read part, it is unnecessary.

Let $\tilde{m}$ (I rather use another variable , same as Wikipedia a or same as article x) be a candidate divisor of n, chosen such that $1 < \tilde{m} < n$. Using the Euclidean algorithm, one can compute $\gcd(\tilde{m}, n)$ in polynomial time [11].

If $\gcd(\tilde{m}, n) \neq 1$, then we have found a non-trivial divisor of n. In this case, set $m = \gcd(\tilde{m}, n)$ and $t = \frac{n}{\gcd(\tilde{m},n)}$.

Otherwise, if $\gcd(\tilde{m}, n) = 1$, then the equivalence class $[\tilde{m}]$ belongs to the multiplicative group,

$$\left(\tfrac{\mathbb{Z}}{n\mathbb{Z}}\right)^* = \{[a] \in \tfrac{\mathbb{Z}}{n\mathbb{Z}} : \gcd(a, n) = 1\}.$$

In this case, we look for the *order* of $[\tilde{m}]$, namely the smallest positive integer r such that $\tilde{m}^r \equiv 1 \pmod{n}$.

This order r always exists since $\left(\tfrac{\mathbb{Z}}{n\mathbb{Z}}\right)^*$ is a finite abelian group. The efficient computation of r is precisely the step where the quantum part of the algorithm becomes essential.

Once r is obtained, two cases arise. If r is odd, the choice of $\tilde{m}$ is not useful, and the procedure is repeated with a new guess $\tilde{m}$. If r is even, then

$$(\tilde{m}^{r/2} - 1)(\tilde{m}^{r/2} + 1) \equiv 0 \pmod{n}.$$

Since r is minimal, we know that $\tilde{m}^{r/2} \not\equiv 1 \pmod{n}$. Thus, it follows that $\tilde{m}^{r/2} \equiv -1 \pmod{n}$ i.e. $\tilde{m}^{r/2} + 1$ is a multiple of n. Therefore, we can compute $\gcd(\tilde{m}^{r/2} +$

$1, n)$. If $\gcd(\tilde{m}^{r/2}+1, n) \neq 1$, then $\gcd(\tilde{m}^{r/2}+1, n)$ is a non-trivial divisor of n, and the factorization goal is achieved. ERNESTO: This is WRONG. $\tilde{m}^{r/2} \not\equiv 1 \pmod{n}$ doesn't imply that $\tilde{m}^{r/2} \equiv -1 \pmod{n}$. In fact, it is desired that this does not happen, since we would have to try another $\tilde{m}$. If $\tilde{m}^{r/2} \equiv -1 \pmod{n}$ i. e. $\tilde{m}^{r/2}+1$ is a multiple of n, when we compute $\gcd(\tilde{m}^{r/2}+1, n)$ we will obtain n which is not a trivial divisor of n, thus having to try another $\tilde{m}$. Sergio: After mull it over my answer is: You are right! I mean I was really convinced that the wikipedia is the ultimate truth, but it is not the case!!!! Concrete counterexample: $n = 15$,$a = 2$. Then $2^4 \equiv 1 \pmod{15}$ so $r = 4$. But $2^{r/2} = 2^2 = 4 \not\equiv \pm 1 \pmod{15}$, even though $(4-1)(4+1) = 3 \cdot 5 \equiv 0 \pmod{15}$. Here $\gcd(4-1, 15) = 3$ and $\gcd(4+1, 15) = 5$ give the nontrivial factors. No joke, we could think of reporting the mistake DAVID: Sergio, I am afraid the Wikipedia is never wrong... Explicitly "At this point, it may or may not be the case that $N|a^{r/2}-1$..."

Otherwise, the procedure is restarted with a new choice of $\tilde{m}$.

QUANTUM FOURIER TRANSFORM

PRIME FACTORIZATION

As discussed above, given x and n, our goal is to find the minimum integer r such that $x^r \equiv 1 \pmod{n}$ Sergio: We can move to other notations but we must be consistent between sections i.e. $x^r \equiv 1 \leftrightarrow \tilde{m}^2 \equiv 1$. The first thing to do is find $q = 2^t$, a power of 2 for certain $t \in \mathbb{N}$, such that $n^2 < q < 2n^2$. Sergio: Here I would add the comment on: we can discard n from being a power of 2 since there are many efficient classical algorithms to factorize n in these cases. Now, starting from the state given by

$$|0\rangle^{\otimes t} \otimes |0\rangle\,,$$

we apply a Hadamard gate to each of the first t qubits. This yields

$$\left(\frac{|0\rangle+|1\rangle}{\sqrt{2}}\right)^{\otimes t} \otimes |0\rangle = \frac{1}{\sqrt{2^t}} \sum_{a=0}^{2^t-1} |a\rangle \otimes |0\rangle = \frac{1}{\sqrt{q}} \sum_{a=0}^{q-1} |a\rangle\,|0\rangle\,,$$

where the tensorial product has expanded into a sum of all possible 2^t binary strings and we have identified each with an integer $a \in \{0, \ldots, 2^m-1\}$. Next, we transform $|0\rangle$ to $|x^a \pmod{n}\rangle$ using *modular exponentiation*. Sergio: Which a are we talking about?? since $a \in \{0, \ldots, 2^m-1\}$. Should be something like: For each a one may compute $|a\rangle\,|0\rangle \to |a\rangle\,|x^a(\mathrm{mod} n)\rangle$ using modular exponentiation This algorithm works as follows. We decompose a as a binary string, i.e., $a = \sum_{i=0}^{l-1} a_i 2^i$ with $a_i \in \{0, 1\}$. Thus

$$x^a(\mathrm{mod} n) = x^{\sum_{i=0}^{l-1} a_i 2^i}(\mathrm{mod} n) = \prod_{i=0}^{l-1} x^{a_i 2^i}(\mathrm{mod} n)$$

$$= \prod_{i=0}^{l-1} \left[x^{2^i}(\mathrm{mod} n)\right]^{a_i}\,,$$

and only when $a_i = 1$, the term is taken into account. Our state is now

$$\frac{1}{\sqrt{q}}\,|a\rangle\,|x^a(\mathrm{mod} n)\rangle$$

Sergio: Maybe I am misunderstanding something but the state is

$$\frac{1}{\sqrt{q}} \sum_{a=0}^{q-1} |a\rangle\,|x^a(\mathrm{mod} n)\rangle\,,$$

Absolutely! Forgot the sum :(and we can perform a quantum Fourier transform to the first register $|a\rangle$ yielding

$$\frac{1}{\sqrt{q}} \sum_{a=0}^{q-1} \sum_{c=0}^{q-1} e^{2\pi i a \frac{c}{q}}\,|c\rangle\,|x^a(\mathrm{mod} n)\rangle\,.$$

Now we proceed with a projective measurement into a general state $|c\rangle\,|x^k(\mathrm{mod} n)\rangle$ and compute the corresponding probability Sergio: Do not repeat the dummy index i.e. $|c\rangle\,|x^k(\mathrm{mod} n)\rangle \to |\tilde{c}\rangle\,|x^k(\mathrm{mod} n)\rangle$, or any other letter but c. This is

$$\mathbb{P}(c, x^k(\mathrm{mod} n)) = \left| \frac{1}{q} \sum_{\substack{a=0 \\ x^a \equiv x^k (\bmod n)}}^{q-1} e^{2\pi i a \frac{c}{q}} \right|^2.$$

Without losing focus on our goal, we can introduce r into the equation Sergio: I don't like this sentence, keep it clean, maybe "Now we introduce r into the equation as the following"... or even remove the sentence entirely. By definition $x^{r+a} \equiv x^a(\mathrm{mod} n)$, so $x^a(\mathrm{mod} n)$ is periodic with period r. This means that the sum can be reinterpreted as a sum over $a \equiv k(\mathrm{mod} r)$ Sergio:It is true, but is more natural to say: Notice that some terms in the sum might be repeated since we can find $|x^i(\mathrm{mod} n)\rangle = |x^j(\mathrm{mod} n)\rangle$ for $i \neq j$. This happens when

$$x^i \equiv x^j(\mathrm{mod} n) \implies i = br + j$$

where we have supposed $i > j$, $b, r \in \mathbb{Z}$ and r satisfies $x^r \equiv 1(\mathrm{mod} n)$ i.e. r is the order of x in $\left(\frac{\mathbb{Z}}{n\mathbb{Z}}\right)^*$. Therefore, $a = br + k$ for some $k \in \mathbb{Z}$ and the probability is reexpressed as

$$\mathbb{P}(c, x^k(\mathrm{mod} n)) = \left| \frac{1}{q} \sum_{b=0}^{\lfloor \frac{q-1-k}{r} \rfloor} e^{2\pi i \frac{c}{q}(br+k)} \right|^2 = \left| \frac{1}{q} \sum_{b=0}^{\lfloor \frac{q-1-k}{r} \rfloor} e^{2\pi i b \frac{rc}{q}} \right|^2.$$

Getting rid of the global unitary factor $e^{2\pi i\frac{c}{q}k}$ in this step looks more logical, then we'll also get rid of the factor $e^{2\pi i b d}$ afterwards. By Euclidean division, $rc = dq + \{rc\}_q$, where $d \in \mathbb{Z}$ and $\{rc\}_q$ is the residue taken in the interval $-\frac{q}{2} < \{rc\}_q \leq \frac{q}{2}$. Then, leaving out the unitary phases, we are left with

$$\mathbb{P}(c, x^k(\mathrm{mod} n)) = \left|\frac{1}{q}\sum_{b=0}^{\lfloor\frac{q-1-k}{r}\rfloor} e^{2\pi i b(\frac{\{rc\}_q}{q}+d)}\right|^2$$

$$= \left|\frac{1}{q}\sum_{b=0}^{\lfloor\frac{q-1-k}{r}\rfloor} e^{2\pi i b\frac{\{rc\}_q}{q}}\right|^2 .$$

We can convert the sum into an integral (I really don't understand this step)[8] (Sergio: Don't worry I don't think Shor understands it either;)) as

$$\frac{1}{q}\int_0^{\lfloor\frac{q-1-k}{r}\rfloor} db\, e^{2\pi i b\frac{\{rc\}_q}{q}} + O\left(\frac{\lfloor (q-1-k)/r\rfloor}{q}\left(e^{2\pi i\frac{\{rc\}_q}{q}}-1\right)\right),$$

where the error term can be bounded by $O(\frac{1}{q})$ if $|\{rc\}_q| \leq \frac{r}{2}$. Under the same condition, the integral term can be shown to be large.

Sergio: I would skip the integral computation at this point (the exact computation is irrelevant), the key is that under some assumptions, an error of $O(\frac{1}{q})$ and assuming $\frac{-r}{2} \leq \{rc\}_q \leq \frac{r}{2}$ it can be shown that

$$\mathbb{P}(c, x^k(\mathrm{mod} n)) \approx \frac{\left|\sin\left(\pi\frac{\{rc\}_q}{r}\right)\right|^2}{\pi^2\left|\{rc\}_q\right|^2}.$$

First, the variable change $u = \frac{rb}{q}$ enables one to express the integral as

$$\frac{1}{r}\int_0^{\frac{r}{q}\lfloor\frac{q-1-k}{r}\rfloor} du\, e^{2\pi i\frac{\{rc\}_q}{r}u} \approx \frac{1}{r}\int_0^1 du\, e^{2\pi i\frac{\{rc\}_q}{r}u},$$

where we assume an error $O(\frac{1}{q})$ in approximating $\frac{r}{q}\lfloor\frac{q-1-k}{r}\rfloor \approx 1$, since $k < r$. The remaining integral can be directly computed as

$$\frac{1}{r}\int_0^1 du\, e^{2\pi i\frac{\{rc\}_q}{r}u} = \frac{e^{2\pi i\frac{\{rc\}_q}{r}}-1}{2\pi i\{rc\}_q} = e^{\pi i\frac{\{rc\}_q}{r}}\frac{\sin\left(\pi\frac{\{rc\}_q}{r}\right)}{\pi\{rc\}_q}.$$

Thus, the probability is given by

$$\mathbb{P}(c, x^k(\mathrm{mod} n)) \approx \frac{\left|\sin\left(\pi\frac{\{rc\}_q}{r}\right)\right|^2}{\pi^2\left|\{rc\}_q\right|^2}.$$

Under the assumption that $\frac{-r}{2} \leq \{rc\}_q \leq \frac{r}{2}$, it is minimized on the boundary of the interval and one obtains an asymptotic lower bound for the probability, which yields $\frac{4}{\pi^2 r^2} \approx 0.405 r^{-2} > \frac{1}{3}r^{-2}$. Therefore the probability will be, at least, $\frac{1}{3r^2}$ as long as $|\{rc\}_q| \leq \frac{r}{2}$. Rearranging this condition by the definition of $\{rc\}_q$ we obtain

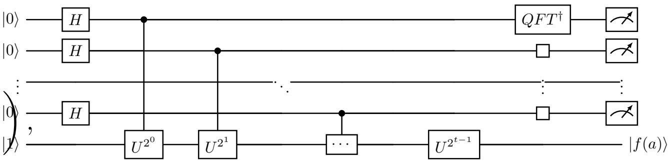
$$\left|\frac{c}{q}-\frac{d}{r}\right| \leq \frac{1}{2q}.$$

In this equation we know both c (we measure it) and q, since its the dimension of our QFT.

TBC

FIGURE

This figure must be polished, but it is essentially Shor's algorithm



CIRCUIT IMPLEMENTATION

CONCLUSIONS AND FUTURE WORK

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[13] This class is called **BPP**, which stands for bounded-error probabilistic polynomial time.